CSA06507-Generative AI

Lab Experiment's

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Unit 2: Lab Programs for Prompting

Experiment 11: Comparative Analysis of Zero-shot, One-shot and Few-shot Prompting

Aim:

To generate responses for the same task using Zero-shot, One-shot, and Few-shot prompting techniques and compare their quality, relevance, accuracy, and consistency.

Algorithm

1. Start the program.
2. Select a common task for generating responses.
3. Design a Zero-shot prompt without providing examples.
4. Design a One-shot prompt by providing one example.
5. Design a Few-shot prompt by providing multiple examples.
6. Generate responses using all three prompting techniques.
7. Display the generated responses.
8. Compare the responses based on quality, relevance, accuracy, and consistency.
9. Display the comparative analysis.
10. Stop the program.

Python Program

Python

Run

Comparative Analysis of Prompting Techniques# Common task:

Classify customer feedback

Zero-shot Prompt

zero_shot = """

Classify the following feedback as Positive or Negative.

Feedback: The product is excellent and very useful.

"""

One-shot Prompt

one_shot = """

Example:

Feedback: The service is very bad.

Sentiment: Negative

Now classify:

Feedback: The product is excellent and very useful.

"""

Few-shot Prompt

few_shot = """

Example 1:

Feedback: I love this product.

Sentiment: Positive

Example 2:

Feedback: The quality is very poor.

Sentiment: Negative

Example 3:

Feedback: The experience was wonderful.

Sentiment: Positive

Now classify:

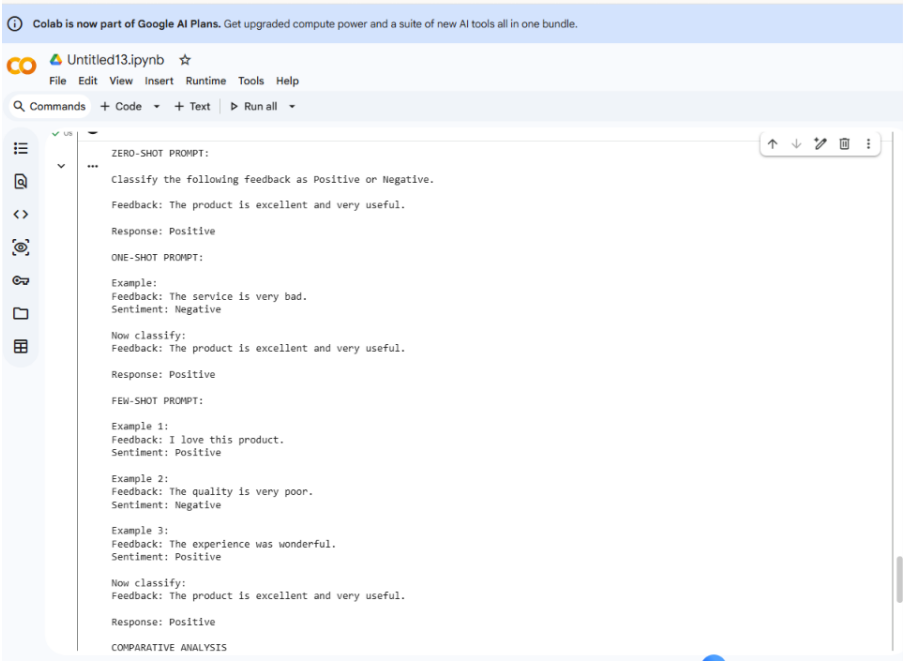
Input

Common task:

Plain text

Feedback: The product is excellent and very useful.

Output



Comparative Analysis

Criteria	**Zero-shot**	**One-shot**	**Few-shot**
Examples	No examples	One example	Multiple examples
Quality	Basic	Improved	More structured
Relevance	Good	Good	Very good
Accuracy	Depends on task	Improved with example	Often improved with examples
Consistency	May vary	Better guidance	More consistent when examples are clear

Explanation

1. Zero-shot Prompting:

Zero-shot prompting generates responses without providing any examples. The model understands the task directly from the instruction.

2. One-shot Prompting:

One-shot prompting provides one example to guide the model in understanding the expected response format.

3. Few-shot Prompting:

Few-shot prompting provides multiple examples to help the model understand the task and generate more consistent responses.

Result

Thus, the comparative analysis of Zero-shot, One-shot, and Few-shot prompting was successfully implemented in Google Colab. All three prompting techniques generated the expected sentiment classification. Their quality, relevance, accuracy, and consistency were compared. Few-shot prompting provides more examples and better guidance, while zero-shot prompting requires no examples.